EXPLAINED SIMPLY

OptiChain

The whole project in plain language: the full story, every term (from stationarity and the ADF test to ARCH and GARCH), every model, and the reason behind every choice.

Saved vs the official method	Over a year (estimate)	Stock on shelves	Best forecaster
€708	**≈ €6,100**	**−45%**	**LightGBM**
in 6 weeks · 16% cheaper	if the test weeks are typical	much less stock gathering dust	error 57.1% vs 63.7%

Contents

Part 1 · The full story, start to finish

Part 2 · Every term explained

Part 3 · Every model explained

Part 4 · Why we compare ten models

Part 5 · The short recap

PART 1

The full story, start to finish

1 · The problem we are trying to solve

Imagine you run a chain of 67 shops. Every shop sells some products, and every Monday morning you place an order with a central warehouse to refill the shelves. The catch is that the delivery takes two weeks: what you order today only reaches the shelf two Mondays from now. So every order is a bet on the future. You must decide today how much customers will want in two weeks' time.

Two things can go wrong. If you order too little, the shelf goes empty, a customer comes in, finds nothing and leaves. You lose that sale for good, because the customer does not wait. The competition puts a price of **€1 on every missed sale**. If you order too much, the extra items sit on the shelf. Holding stock costs money (space, capital, risk of damage), and the competition charges **€0.20 for every item still on the shelf at the end of each week**.

A missed sale is five times more expensive than keeping one item for a week. That one fact shapes the whole project: it is better to keep a little extra stock than to run out, but extra stock is not free either. The goal is to find the balance, ordering the amount that keeps the **total cost** (missed sales plus leftover stock) as low as possible, for every product in every shop, every week.

2 · The data and why we chose it

The data comes from a real competition called the **VN2 Inventory Planning Challenge**, run in 2025 by a supply-chain expert, Nicolas Vandeput. About 180 teams entered. The sales numbers are real weekly sales from an anonymous retail chain, covering April 2021 to June 2024: 165 weeks in total.

The chain has 67 shops and 297 products, but not every shop sells every product. When you count the combinations that actually exist, there are **599 store-products**. A store-product is simply one product in one shop. "Product 124 in Shop 61" is one store-product, and "Product 124 in Shop 63" is a different one. They are treated separately because the same product can sell very differently in two places.

Shops and products appear only as numbers because the company chose to stay anonymous. We never invented real names. On the dashboard five example products get descriptive nicknames such as "The Christmas Rocket" (it sells almost three times more in December), but the nickname only describes how the product behaves, and its real code is always shown next to it.

The project first used Walmart sales data (the M5 dataset). We switched because that data is older (2011-2016) and has no information about stock, costs or delivery times, so those had to be made up. VN2 gives us the **real stock levels, the real costs, the real two-week delivery rule and an official score**, so we can measure our work against 180 real teams.

We split the 165 weeks into two parts. The first **157 weeks** are history, used for learning. The last **8 weeks** are kept hidden and used only once, at the very end, as the final exam. If we tuned anything on those 8 weeks, the final score would be dishonestly good.

3 · Checking the data before trusting it

Before building anything, we checked that the data was healthy: no empty cells, no negative sales, no fractions of an item, and all nine weekly files agreed with each other. Everything passed. This matters because any problem here would quietly corrupt every result later.

Two facts stood out. First, **43% of all weeks have zero sales**: most products are slow sellers, about one unit a week, and many weeks they sell nothing at all. That makes forecasting hard, because there is very little signal in such small numbers. Second, **about one week in ten was flagged as out of stock**, meaning the shelf was empty at the end of that week. That turns out to be crucial.

4 · Looking at the data with charts (EDA)

EDA (exploratory data analysis) means studying the data with charts and simple statistics before choosing any method. Each chart answers one question.

The total sales line graph adds up all 599 store-products week by week. It shows a steady level of about 1,760 units a week with a large spike every December: a strong **Christmas season**. **The seasonal profile** puts numbers on it: Christmas weeks sell about **2.4 times** a normal week.

The histograms show that the middle product sells about 1.1 units a week and three quarters sell fewer than 2. **The Pareto curve** shows that a small group of products makes most of the sales, so mistakes on big sellers matter most. **The scatter plots** show that small sellers are the most irregular, and that this week's total is strongly linked to last week's. **The QQ plots** show that sales do not follow a bell curve. **The stockout charts** show that empty-shelf weeks range from 0% in the best store to 59% in the worst.

From this we learned four things that shaped every later choice: demand is slow and often zero; there is a strong yearly season shared by all shops; empty shelves are common and distort the data; and products share patterns, so learning from all of them together should help.

5 · Hidden missing values: the most important cleaning step

The audit said there are no missing values: every cell has a number. But some of those numbers do not mean what they seem to mean.

The first problem is products before their launch. Many products show zeros in their first weeks, not because nobody wanted them but because they were not on sale in that shop yet. Every week before a product's first sale is treated as missing.

The second, bigger problem is empty-shelf weeks. When the shelf was empty, sales were zero, but customers may well have wanted the product; they just could not buy it. This is called **censored demand**. We found that **100% of out-of-stock weeks recorded exactly zero sales**: those zeros measure supply, not demand.

We set these weeks aside, and call what remains **clean demand**. Filling them with zero would teach the models that people want less than they do, so they would order too little and cause even more empty shelves. Filling them with a guess would be invented information. So each method simply skips these weeks.

We also sorted every product into one of four **demand patterns**: steady (sells almost every week, similar amounts), spiky (sells most weeks, amounts jump), occasional (many weeks with no sales) and unpredictable (irregular weeks and amounts). Most of the 599 are occasional or unpredictable.

6 · The classical time-series analysis

A time series is a list of numbers in time order, like weekly sales. Statisticians have a classic recipe for building a forecasting model from one, the **Box-Jenkins method**, and we followed it step by step on the network total, on one representative product (Store 61 · Product 124), and automatically on all 599.

Is it stable? The ADF test said yes for the total and for 94% of products. **Does it need transforming?** No, it was already stable. **Trend, season and noise:** STL found a seasonal strength of 0.93 out of 1, so the yearly pattern dominates. **Which past weeks matter?** The ACF and PACF charts showed the last one to five weeks and the same week last year. **Which model type?** The pattern pointed to SARIMA. **Which exact version?** Out of 27 candidates, AIC chose **SARIMA(2, 0, 0)x(1, 0, 0)[52]**: this week depends on the last two weeks and on the same week last year.

What does the model leave behind? The ARCH test found calm and wild periods (Christmas is wild), so we added **GARCH**, after which no pattern remained: the leftovers are pure random noise. **Does it predict unseen weeks?** It forecast the 8 hidden weeks with 14% error (against 20% for copying last year), and all 8 real weeks fell inside its predicted range. Across all 599 products, 91% of the individual ARIMA models left pure noise behind.

Part 2 explains every one of these terms in detail.

7 · The ten forecasting models

We did not pick one method and hope for the best. We built ten, from the simplest guess ("next week = last week") to machine learning, and compared them fairly so that the final choice is proven rather than assumed. Part 3 explains each one; Part 4 explains why comparing them matters.

8 · Testing the models fairly (the backtest)

We went back in time to **30 different past weeks**. At each one, every model could see only the data up to that week and had to forecast the next three weeks; then we compared the forecasts with what actually sold. Letting a model see the future would be cheating (leakage). Three weeks because an order placed now arrives in two weeks, so the planner needs this week, next week and the arrival week.

LightGBM won clearly, with an error of 57.1%. The ensemble came second (59.6%), the official benchmark scored 63.7%, ARIMA 65.0%, and "last week" was worst (75.6%). The errors look large because most products sell one or two units a week: guessing 2 when 1 sells is already a 100% error. What matters is the comparison, and later the cost.

9 · Knowing how uncertain a forecast is

A forecast is just the most likely value. Ordering exactly that would run out about half the time. So we modelled how spread out demand is around each forecast, with a **negative binomial distribution** (a shape for counts with many zeros). When it said "80% chance demand stays below X", demand stayed below X about 85% of the time: slightly cautious, and corrected in the next steps.

10 · The shop simulator

We wrote a simulator that plays the shops week by week with the exact competition rules. Fed the organiser's own benchmark orders, it produced exactly **€4,334.0**, the official number, to the cent. So every score in the project is directly comparable with the real competition.

11 · Deciding how much to order

We used the **Monte Carlo newsvendor**: imagine 1,000 possible futures, see how much stock will be left when the new order arrives, and order enough to cover demand in a chosen share of those futures (the **service level**). The textbook share is 83%, but replaying 18 past weeks showed **75%** was cheaper, because our uncertainty estimate was slightly cautious and leftover stock is not wasted (it sells the next week).

12 · The final exam: the eight hidden weeks

Replaying the competition exactly, with nothing tuned on these weeks, our plan (the ensemble at 75%) cost **€3,626**, against **€4,334** for the official method: **€708 (16%) cheaper**, roughly €6,100 a year. The real winner scored €3,763.

The saving comes from keeping about **45% less stock** on the shelves, in exchange for somewhat more missed sales. Because a leftover costs €0.20 every week while a missed sale costs €1 once, holding much less stock saves more than the extra missed sales cost.

Two honest caveats: we built this after the competition, knowing which methods worked, so "we beat the official method" is fair but "we beat the winner" is not fully fair; and on these 8 weeks the simple benchmark made slightly more accurate forecasts, yet our orders still cost less.

13 · What to order next

The data ends on 3 June 2024, so the system treats 10 June 2024 as "today" and recommends ordering **1,162 units across 271 store-products**, arriving on 24 June. It flags 19 products likely to run out next week, which is already too late to fix because any order takes two weeks. It cannot plan for today's real date: there is no data after June 2024.

14 · The lessons

Cleaning the data mattered as much as the models. Seasonality is strong and best learned across all shops. The classical statistics check out. Machine learning predicts best because it learns from all products together. A good forecast is not enough: the ordering rule must handle uncertainty and real costs. And testing decisions with simulated costs beats trusting a formula.

PART 2

Every term explained

Each term has three parts: **what it is** in plain words, a small **example**, and **why we used it** in this project. The terms are grouped by topic, in the order you meet them.

The business terms

Store-product

What it is	One product in one shop. It is the unit of every decision in this project.
Example	Product 124 in Shop 61 and Product 124 in Shop 63 are two different store-products.
Why we used it	Each one sells differently, so each gets its own forecast and its own order. There are 599.

Lead time

What it is	The time between placing an order and the stock arriving on the shelf.
Example	Order on Monday 15 April, the stock arrives on Monday 29 April: a lead time of two weeks.
Why we used it	Because of it, every order must be based on a forecast of the future, not on what is on the shelf today.

Holding cost

What it is	The cost of keeping one unsold item on the shelf for one week. In VN2 it is €0.20.
Example	10 items left at the end of a week cost €2.
Why we used it	It is half of the total cost we try to minimise.

Shortage cost (lost sale)

What it is	The cost of a customer wanting an item when the shelf is empty. In VN2 it is €1 per unit, and the sale is lost (the customer does not wait).
Example	5 customers find an empty shelf: €5.
Why we used it	It is the other half of the total cost, and five times more expensive than holding.

Week 0

What it is	8 April 2024, the last week of history before the competition weeks.
Example	Week 1 is 15 April 2024, week 8 is 3 June 2024.
Why we used it	Everything up to week 0 is for learning; weeks 1-8 are the final test.

The data terms

Censored demand

What it is	Sales that are lower than real demand because the shelf ran out. The real demand is hidden, or "censored", behind the recorded number.
Example	10 customers wanted an item, but only 6 were on the shelf: the record says 6, the real demand was 10. In our data the shelf was empty all week, so the record says 0.
Why we used it	We found that every out-of-stock week shows zero sales, so we set those weeks aside instead of treating them as demand.

Clean demand

What it is	Sales with the untrustworthy weeks removed: weeks before launch and out-of-stock weeks.
Example	A product launched in week 5 and out of stock in week 20: weeks 1-4 and 20 are removed.
Why we used it	All models learn only from clean demand, so they are not dragged down by false zeros.

NaN (missing value)

What it is	"Not a number": an empty value used to mark a week we do not trust.
Example	An out-of-stock week becomes NaN instead of 0.
Why we used it	Models skip NaN weeks instead of treating them as real zeros.

Demand pattern (ADI and CV^2)

What it is	A label for how a product sells. ADI is the average number of weeks between sales; CV^2 measures how much the size of a sale varies. Together they give four patterns: steady, spiky, occasional, unpredictable.
Example	A product that sells 2 units every week is steady; one that sells 0, 0, 7, 0, 1 is unpredictable.
Why we used it	It tells us which products are hard to forecast and which special methods (Croston, TSB) to include.

EDA (exploratory data analysis)

What it is	Studying the data with charts and simple statistics before building models.
Example	Drawing total sales over time and spotting the Christmas peak.
Why we used it	It decided many later choices: the season, slow demand, censoring.

Seasonality and seasonal factor

What it is	A pattern that repeats at the same time every year. The seasonal factor of a week is its average sales divided by an average week.
Example	Christmas weeks have a factor of about 2.4: they sell 2.4 times a normal week.
Why we used it	Forecasts must know the time of year, so most models use these factors.

The statistics terms: distributions and tests

Hypothesis test and p-value

What it is	A test starts from an assumption (the "null hypothesis", H0) and asks how likely the data would be if it were true. The p-value is that probability. If it is below 0.05 (5%), we reject the assumption.
Example	Assumption: "the coin is fair". Ten heads in a row has a p-value of about 0.001, so we reject it: the coin is probably not fair.
Why we used it	Every statistical test in the project (ADF, KPSS, Ljung-Box, Jarque-Bera, ARCH-LM) gives a p-value that we read this way.

Normal distribution (bell curve)

What it is	The familiar symmetric bell shape, where most values are near the average and extreme values are rare.
Example	People's heights roughly follow a bell curve.
Why we used it	Many classical methods assume it. We checked whether our data follows it (it does not), which changed how we model uncertainty.

QQ plot

What it is	A chart that compares the data with a bell curve. If the points lie on a straight line, the data is normal; if they bend away, it is not.
Example	Our sales bend sharply upwards at the end: a few very big weeks.
Why we used it	It showed that sales are not normal, so we model uncertainty with a count distribution instead.

Skewness and kurtosis

What it is	Skewness measures how lopsided the data is; kurtosis measures how heavy the extremes are.
Example	Most weeks sell 0-2 units but a few sell 30: strongly skewed, heavy extremes.
Why we used it	Both confirmed that sales are far from a bell curve.

Jarque-Bera test

What it is	A test of normality based on skewness and kurtosis. H0: the data is normal. p below 0.05 means not normal.
Example	Our residuals gave p = 0.000: not normal.
Why we used it	It checks whether normal-based prediction ranges would be trustworthy.

Poisson and negative binomial distributions

What it is	Shapes for counts (0, 1, 2, ...). Poisson is for pure random counts; the negative binomial allows more spread, for counts that are sometimes zero and sometimes spike.
Example	A product averaging 2 a week might sell 0, 1, 2, 5 or 9: the negative binomial describes that spread.
Why we used it	We use it to describe the uncertainty around each forecast, because weekly sales are small counts with many zeros.

The statistics terms: stability and seasons

Stationarity (a stationary series)

What it is	A series is stationary when its average level and its spread stay the same over time. It moves around, but around a stable centre, and does not drift away.
Example	Daily temperature differences from the yearly average are roughly stationary; a company's growing revenue is not (it drifts upward).
Why we used it	AR, MA, ARMA and ARIMA models only work properly on stationary data, so stationarity is the first thing to check.

Unit root and random walk

What it is	A random walk is a series where each value is the previous value plus a random step, so it can wander anywhere and never returns to a centre. Having a "unit root" means behaving like this.
Example	A drunk person's walk: each step starts from wherever the last one ended.
Why we used it	A unit root means the series is not stationary, and the ADF test is designed to detect one.

ADF test (Augmented Dickey-Fuller)

What it is	The standard test for stationarity. Its assumption (H0) is that the series has a unit root, i.e. it is NOT stationary. A p-value below 0.05 rejects that, so the series is stationary. "Augmented" means it also accounts for the effect of recent weeks.
Example	Network total: ADF p-value 0.006, below 0.05, so the series is stationary.
Why we used it	To check that the classical models can be used, and to decide whether differencing is needed. 94% of our products passed.

KPSS test

What it is	A second stationarity test that asks the opposite question: its H0 is that the series IS stationary. p below 0.05 means not stationary.
Example	If ADF says stationary and KPSS agrees, we are confident; if they disagree, we look closer.
Why we used it	A cross-check on ADF, to avoid a wrong conclusion from one test alone.

Transformation: log and differencing

What it is	Ways to make a series stationary. The log shrinks big values (calming a spread that grows with the level). Differencing replaces each value by its change from the previous week. A seasonal difference subtracts the same week last year.
Example	Sales of 100, 110, 125 become differences of 10 and 15.
Why we used it	We tested them all; none was needed because the data was already stationary (d = 0).

d (order of differencing)

What it is	How many times the series is differenced before modelling. d = 0 means use it as it is.
Example	d = 1 means model the weekly changes instead of the levels.
Why we used it	The ADF test chose d = 0 for our series.

Trend

What it is	The slow, long-term movement of a series, up or down.
Example	Sales gently rising over three years.
Why we used it	One of the three parts STL separates.

STL decomposition

What it is	Seasonal-Trend decomposition using LOESS: it splits a series into trend + season + remainder (noise), and gives each part a strength from 0 to 1.
Example	Our total: seasonal strength 0.93, the yearly pattern dominates.
Why we used it	To measure how seasonal the data is, which led to taking the season from the whole network.

The statistics terms: models and choosing them

Lag

What it is	A past value of the series. Lag 1 is last week, lag 52 is the same week last year.
Example	To predict this week, we might look at lag 1 (last week) and lag 52 (a year ago).
Why we used it	We found lags 1-5 and 52 matter, and gave them to the machine-learning model as inputs.

ACF (autocorrelation function)

What it is	A chart of how strongly the series is linked to itself k weeks earlier, for k = 1, 2, 3, ... Bars outside the shaded band are significant.
Example	A tall bar at lag 1 means this week is a lot like last week.
Why we used it	It shows which past weeks matter and helps identify the model type.

PACF (partial autocorrelation)

What it is	Like the ACF, but it keeps only the direct link with k weeks ago, removing the influence of the weeks in between.
Example	If this week links to two weeks ago only because both link to last week, the PACF at lag 2 is small.
Why we used it	Together with the ACF, it tells AR and MA patterns apart.

AR model (autoregressive)

What it is	This week is predicted from a weighted sum of the last few weeks' values. AR(p) uses p past weeks.
Example	AR(2): this week = 0.32 × last week + 0.15 × two weeks ago + a constant.
Why we used it	Our total shows an AR pattern.

MA model (moving average)

What it is	This week is predicted from the last few surprises (forecast errors). MA(q) uses q past surprises.
Example	MA(1): if last week was 10 units higher than expected, part of that surprise carries into this week.
Why we used it	Our representative product was best described by MA(1).

ARMA, ARIMA, SARIMA

What it is	ARMA combines AR and MA. ARIMA is ARMA applied after differencing (the I is for "integrated"). SARIMA adds a seasonal part that links each week to the same week in earlier years. They are written (p, d, q) for the regular part and (P, D, Q)[52] for the seasonal part.
Example	SARIMA(2, 0, 0)x(1, 0, 0)[52]: two past weeks (p = 2), no differencing, and one seasonal AR term at lag 52.
Why we used it	They are the classical forecasting models. SARIMA describes our network total; ARIMA is one of our ten forecasting models.

Likelihood

What it is	How probable the observed data is under a given model. A better-fitting model has a higher likelihood.
Example	A model that predicts the past weeks closely has a high likelihood.
Why we used it	AIC and BIC are built from it.

AIC and BIC

What it is	Scores for choosing between models. Both reward a good fit (likelihood) and punish complexity (number of parameters); BIC punishes more strictly. Lower is better; a difference below 2 means the models are about equal.
Example	Of 27 candidates, AIC chose SARIMA(2,0,0)(1,0,0)[52]; BIC chose a version with one parameter fewer.
Why we used it	To pick the best model objectively, without over-complicating it (overfitting).

Overfitting

What it is	When a model learns the noise of the past instead of the real pattern, so it looks great on old data but predicts the future badly.
Example	Memorising last year's exact random spikes.
Why we used it	AIC/BIC penalties and the fair backtest protect against it.

Kalman filter

What it is	The algorithm that fits ARIMA-type models step by step through time. It can simply skip missing weeks.
Example	An out-of-stock week is skipped instead of being filled in.
Why we used it	It lets ARIMA learn from clean demand with gaps.

The statistics terms: checking the leftovers

Residuals

What it is	What a model could not explain: actual value minus the model's prediction.
Example	The model predicted 1,800, and 1,900 sold: the residual is +100.
Why we used it	If residuals still show a pattern, the model missed something.

White noise

What it is	A series with no pattern at all: no links between weeks and a constant spread. Pure randomness.
Example	The results of rolling a die every week.
Why we used it	A good model leaves white-noise residuals: everything predictable has been captured.

Ljung-Box test

What it is	Tests whether residuals are white noise by checking many lags at once. H0: no autocorrelation. p above 0.05 means white noise (good).
Example	After GARCH, our residuals gave p = 0.25 at lag 13: white noise.
Why we used it	To confirm the models leave no pattern behind; also used on real forecast errors.

Volatility and volatility clustering

What it is	Volatility is the size of the ups and downs. Clustering means calm periods and wild periods come in runs: after a wild week, more wild weeks tend to follow.
Example	Around Christmas, forecast errors are much bigger for several weeks, then calm down.
Why we used it	Our residuals showed clustering, which led to ARCH and GARCH.

ARCH effect and the ARCH-LM test

What it is	ARCH (Autoregressive Conditional Heteroskedasticity) means the size of the errors depends on the size of recent errors. The ARCH-LM test checks for it. H0: constant spread. p below 0.05 means an ARCH effect is present.
Example	Our SARIMA residuals gave p = 0.027: there is an ARCH effect.
Why we used it	To detect volatility clustering, so we know whether a GARCH model is needed.

GARCH(1,1)

What it is	Generalised ARCH: a model of how volatility changes over time. Today's volatility = a base level (ω) + a reaction to yesterday's shock (α) + a memory of yesterday's volatility (β). α + β is the persistence: how long wild periods last.
Example	Our total: ω = 0.35, α = 0.18, β = 0.45.
Why we used it	After GARCH, the residuals pass Ljung-Box and ARCH-LM: no pattern and no clustering left. It shows the leftover is pure noise once changing volatility is accounted for.

Standardised residuals

What it is	Residuals divided by their GARCH volatility, so wild and calm periods are put on the same scale.
Example	A +200 error in a wild week and a +50 error in a calm week might both become +1.
Why we used it	These are what we re-test after GARCH.

The forecasting terms

Forecast and horizon

What it is	A forecast is a prediction of future demand. The horizon is how many weeks ahead.
Example	A 3-week horizon predicts this week, next week and the week after.
Why we used it	Orders need 3 weeks: two to wait, one to cover.

Backtest (rolling origin)

What it is	Testing a model by pretending to be in the past: at each of many past dates, it may only use data up to that date, and its forecasts are compared with what really happened.
Example	Standing on 1 October 2023, forecast October; then move to 8 October and repeat.
Why we used it	The fair way to measure accuracy (30 dates).

Leakage

What it is	Accidentally letting a model see information from the future. It makes results look better than reality.
Example	Training on next month's sales and then "predicting" next month.
Why we used it	The whole design avoids it: models only ever see the past.

MAE, WAPE, bias and Score

What it is	MAE: the average miss in units. WAPE: the total miss divided by total sales. Bias: whether forecasts are too high (+) or too low (−) overall. Score = WAPE + |bias|, the official metric of the VN1 forecasting competition.
Example	Forecasts 3, 2, 1 against sales 2, 2, 2: MAE 0.67, WAPE 33%, bias 0%.
Why we used it	To compare models on accuracy and on systematic over- or under-forecasting.

Feature

What it is	An input given to a machine-learning model.
Example	Last week's sales, the time of year, the store.
Why we used it	LightGBM uses about 30 features.

Decision tree and gradient boosting

What it is	A decision tree makes a prediction through a series of yes/no questions. Gradient boosting builds many small trees one after another, each correcting the errors of the previous ones.
Example	Tree 1: "Is it December?". Tree 2 fixes what tree 1 got wrong for slow sellers.
Why we used it	This is how LightGBM works.

Tweedie loss

What it is	The measure of error that LightGBM tries to reduce while learning, designed for data that is never negative and has many exact zeros.
Example	Weekly sales of 0, 0, 1, 0, 3.
Why we used it	It beat the usual alternatives on our data.

Ensemble

What it is	Combining several models, here by averaging their forecasts.
Example	The average of three forecasts of 4, 6 and 5 is 5.
Why we used it	Different models make different mistakes; averaging cancels some of them.

Calibration

What it is	Whether stated probabilities come true as often as claimed.
Example	If "80% sure" is right 80% of the time, the model is calibrated.
Why we used it	We checked it for the uncertainty ranges (they were right about 85% of the time).

The inventory terms

Simulator

What it is	Code that plays the shops week by week with the official rules and adds up the costs.
Example	Start 10 units, demand 12: sell 10, lose 2 sales (€2), end with 0.
Why we used it	To score any ordering plan exactly like the competition did (it reproduced €4,334.0).

Newsvendor problem and critical ratio

What it is	The classic problem of choosing an order size before knowing demand, with a cost for too much and a cost for too little. The best order covers demand with a probability equal to the critical ratio = shortage cost ÷ (shortage cost + holding cost).
Example	With €1 and €0.20: 1 ÷ 1.2 = 0.833, so cover demand 83% of the time.
Why we used it	It is the theory behind our ordering rule.

Service level

What it is	The chance that an order covers the demand of the week it arrives.
Example	75% means covering demand in 3 out of 4 imagined futures.
Why we used it	The main setting of the ordering rule; we tuned it to 75%.

Monte Carlo simulation

What it is	Estimating something by simulating many random scenarios and looking at the results.
Example	Imagine 1,000 possible next weeks of demand and see how often the shelf runs out.
Why we used it	It lets us account for the stock already on its way and the uncertainty of the next two weeks.

Order-up-to rule

What it is	Order enough to bring stock on hand plus stock on its way up to a target, such as N weeks of forecast.
Example	Target 4 weeks of forecast = 40 units; 25 on hand and on the way, so order 15.
Why we used it	It is the official benchmark's rule, kept for comparison.

Safety stock

What it is	Extra stock kept above the expected demand to protect against surprises.
Example	Expecting 10, keeping 13.
Why we used it	The service level decides how much safety stock the plan holds.

Tuning by replay

What it is	Choosing a setting by replaying past weeks with each candidate setting and keeping the cheapest.
Example	Replaying 18 past weeks at service levels from 50% to 96%.
Why we used it	It found 75% was cheaper than the textbook 83%.

PART 3

Every model explained

Ten models were built and compared on the same 30 past weeks (accuracy: error %, lower is better) and on the 8 competition weeks (cost of the orders each model leads to, lower is better). The official benchmark is 63.7% error and €3,751 on our replay of the 8 weeks.

1 · Last week (naive)

How it works	Next week will be exactly like the last normal week.
Small example	Sold 5 last week, so predict 5 for each coming week.
Strengths	Instant reaction to change; trivial to understand.
Weaknesses	Copies every random spike; ignores the season.
Why it is in the comparison	It is the simplest possible guess: any serious method must beat it, or it is not worth its complexity.
How it did	Worst: error 75.6%, orders cost €5,218.

2 · 13-week average

How it works	Average of the last 13 normal weeks (about three months).
Small example	Sales 4, 6, 5, ... over 13 weeks average 5, so predict 5.
Strengths	Smooths out random ups and downs; stable.
Weaknesses	Ignores the season; slow to react when demand really changes.
Why it is in the comparison	A classic simple baseline that shows how much smoothing alone helps.
How it did	Error 69.4%, orders €4,277.

3 · Same week last year (seasonal naive)

How it works	Copy what sold in the same week one year earlier.
Small example	Sold 20 in Christmas week last year, so predict 20 this Christmas.
Strengths	Captures the season automatically.
Weaknesses	Repeats last year's random noise; useless for new products.
Why it is in the comparison	Tests whether the season alone is enough to forecast.
How it did	Error 70.8%, orders €5,468: the season alone is not enough.

4 · Official benchmark (seasonal moving average)

How it works	Removes the season from recent sales, averages the last 13 weeks, then puts the season back for each future week.
Small example	Average 5 per normal week; a Christmas week has factor 2.4, so predict 12.
Strengths	Sensible, simple and seasonal; the competition's own method.
Weaknesses	Same weight for all 13 weeks; one season for all products.
Why it is in the comparison	It is the official yardstick of the competition: beating it is the goal.
How it did	Error 63.7%, orders €3,751.

5 · Exponential smoothing

How it works	A weighted average that trusts recent weeks more: weights fade gradually into the past. The speed of fading is chosen per product. The season is removed first and added back afterwards.
Small example	Weights 30% on last week, 21% on the week before, 15% on the one before, and so on.
Strengths	Reacts faster than a plain average while still smoothing noise.
Weaknesses	Cannot model trends or links to specific past weeks.
Why it is in the comparison	A classic, reliable statistical method that is hard to beat in practice.
How it did	Error 66.1%, orders €3,817.

6 · ARIMA (AIC-selected)

How it works	The Box-Jenkins model of Part 2, one per product: ADF picks the differencing, AIC picks how many past weeks and past surprises to use. The season is removed first.
Small example	AR(1): this week = 0.4 × last week + a constant, then the season is added back.
Strengths	Statistically principled; captures how weeks depend on each other; residuals checked.
Weaknesses	Each product learns alone from little data; struggles with many zeros.
Why it is in the comparison	It is the classical time-series approach, and the subject of the whole Box-Jenkins analysis.
How it did	Error 65.0%, orders €3,757: it beats the official method on cost.

7 · Croston (SBA)

How it works	Built for slow sellers. It separately tracks how big a sale usually is and how many weeks usually pass between sales; the forecast is size ÷ gap, with a small correction (SBA) for a known upward bias.
Small example	A sale of 4 roughly every 4 weeks: about 1 per week.
Strengths	Designed for intermittent demand with many zeros.
Weaknesses	Keeps forecasting the old rate even if the product stops selling.
Why it is in the comparison	Most of our products are occasional or unpredictable sellers, the case Croston was invented for.
How it did	Error 65.6%, orders €3,977.

8 · TSB

How it works	Like Croston, but it tracks the probability of a sale each week (updated every week, sale or not) and the usual size of a sale; forecast = probability × size.
Small example	Sells in 1 week out of 4, 4 units each time: 0.25 × 4 = 1 per week.
Strengths	Notices when a product dies: the probability fades towards zero.
Weaknesses	Still treats each product alone.
Why it is in the comparison	A modern improvement on Croston for products that may stop selling.
How it did	Error 64.3%, orders €4,009.

9 · LightGBM (machine learning)

How it works	One gradient-boosted decision-tree model trained on all 599 products at once. Hundreds of small trees ask questions like "Is it December? Were sales high last month? Was the shelf empty recently?", each tree correcting the previous ones.
Small example	It learns that products like this one, in December, after a strong month, usually sell about 30% more.
Strengths	Learns from all products together, so slow sellers borrow patterns from similar ones; uses many inputs.
Weaknesses	Harder to explain; needs careful testing to avoid overfitting.
Why it is in the comparison	Global machine-learning models like this won the VN2 competition; we need to know if it beats the classics here.
How it did	**Best forecaster**: error 57.1%, orders €3,680.

10 · Ensemble

How it works	The average of three different models: LightGBM, exponential smoothing and the official benchmark.
Small example	Forecasts of 6, 4 and 5 average to 5.
Strengths	Different models make different mistakes, and averaging cancels some of them.
Weaknesses	Not always the most accurate single forecast.
Why it is in the comparison	Combining forecasts is one of the most reliable findings in forecasting research.
How it did	**Cheapest orders**: error 59.6%, orders €3,626: our chosen plan.

PART 4

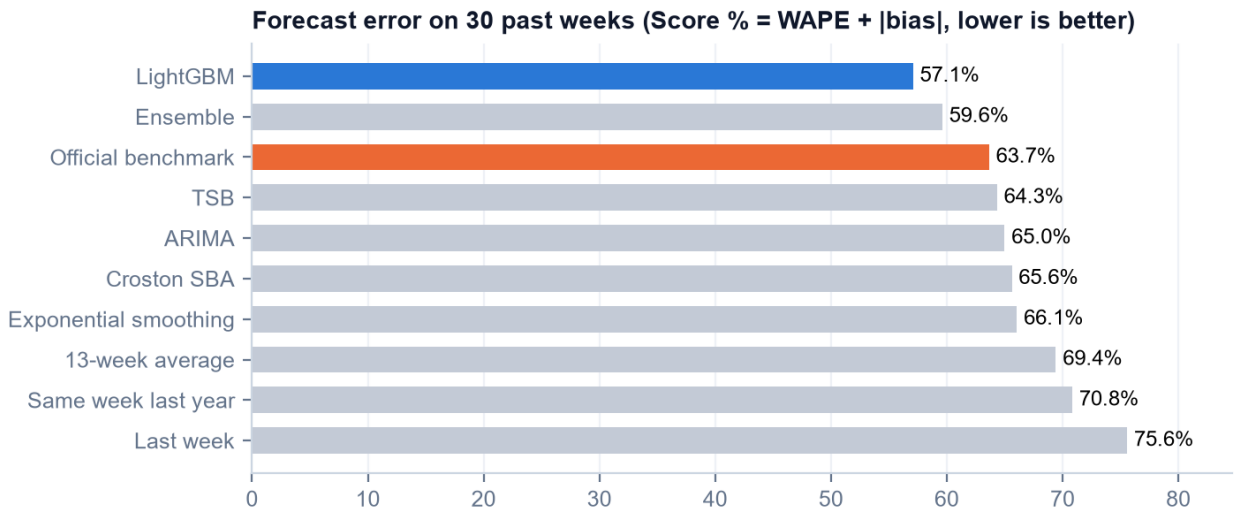
Why we compare ten models

Choosing a single model and trusting it would be guesswork. Comparing ten models on exactly the same data, the same past weeks and the same rules turns the choice into evidence. Each model is in the comparison for a reason:

Group	Models	What the comparison tells us
Simple baselines	Last week, 13-week average, same week last year	How much a method gains over the simplest possible guesses. A complex model that cannot beat them is useless.
The official yardstick	Official benchmark	Whether we beat the competition's own method: the real target.
Classical statistics	Exponential smoothing, ARIMA	Whether the textbook approaches (including the full Box-Jenkins analysis) are good enough on their own.
Slow-seller specialists	Croston, TSB	Whether methods built for intermittent demand help, since most products are slow sellers.
Machine learning	LightGBM	Whether learning across all products together beats per-product methods.
Combination	Ensemble	Whether averaging different models gives safer forecasts and cheaper orders.

What the comparison showed. LightGBM made the most accurate forecasts over 30 past weeks. Every advanced method beat the simple baselines. ARIMA, exponential smoothing and the benchmark were close to each other. The ensemble led to the cheapest orders. Accuracy and cost are related but not identical: what finally matters to the shop is the cost of the orders, so the plan was chosen on cost, tested on past weeks.

Why test on cost and not only on accuracy? The ordering rule turns forecasts into decisions, and small differences in how a model over- or under-predicts can change the cost more than its average accuracy suggests. That is why every model was also played through the official simulator.



Forecast error of the ten models on 30 past weeks (lower is better).

PART 5

The short recap

Topic	In one or two sentences
The problem	Order stock every Monday that arrives two weeks later. Empty shelf: €1 per lost sale. Leftover: €0.20 per item per week. Find the cheapest balance.
The data	Real VN2 competition data: 67 shops, 297 products, 599 store-products, 165 weeks. The last 8 weeks are the final test.
The cleaning	Zeros before a launch and in empty-shelf weeks are not real demand (100% of empty-shelf weeks show zero), so they are set aside.
The charts	A strong Christmas season (2.4× a normal week), slow sellers (about 1 a week), not bell-shaped, many empty shelves.
The statistics	Stationary (ADF), very seasonal (0.93), best classical model SARIMA(2, 0, 0)x(1, 0, 0)[52], leftovers are pure noise after GARCH.
The models	Ten models, from "last week" to machine learning, compared fairly on 30 past weeks.
The best forecaster	LightGBM, error 57.1% against 63.7% for the official benchmark.
The ordering rule	Imagine 1,000 futures and cover demand in 75% of them (tested on past weeks; the textbook says 83%).
The result	€3,626 against €4,334: 16% cheaper, with 45% less stock on the shelves.
The honest limits	Built with hindsight; only 8 test weeks; empty-shelf weeks still understate demand; products are anonymous.